

Supplemental Material

Table S1. Available data period of PM_{2.5} in each prefecture.

Prefecture	Data period	Prefecture	Data period
Hokkaido	2012-2019	Shiga	2014-2019
Aomori	2013-2019	Kyoto	2012-2019
Iwate	2012-2019	Osaka	2012-2019
Miyagi	2012-2019	Hyogo	2012-2019
Akita	2013-2019	Nara	2013-2019
Yamagata	2013-2019	Wakayama	2012-2019
Fukushima	2015-2019	Tottori	2016-2019
Ibaraki	2013-2019	Shimane	2012-2019
Tochigi	2014-2019	Okayama	2013-2019
Gumma	2012-2019	Hiroshima	2012-2019
Saitama	2012-2019	Yamaguchi	2013-2019
Chiba	2012-2019	Tokushima	2012-2019
Tokyo	2012-2019	Kagawa	2013-2019
Kanagawa	2012-2019	Ehime	2012-2019
Niigata	2013-2019	Kochi	2013-2019
Toyama	2013-2019	Fukuoka	2012-2019
Ishikawa	2014-2019	Saga	2013-2019
Fukui	2013-2019	Nagasaki	2015-2019
Yamanashi	2012-2019	Kumamoto	2014-2019
Nagano	2012-2018	Oita	2015-2019
Gifu	2012-2019	Miyazaki	2015-2019
Shizuoka	2012-2019	Kagoshima	2012-2019
Aichi	2012-2019	Okinawa	2015-2019
Mie	2012-2019		

Table S2. Descriptive statistics for out-of-hospital cardiac arrest (OHCA), PM_{2.5}, and temperature in the 47 prefectures of Japan, 2012-2019.

Prefecture	OHCA		PM _{2.5} (µg/m ³)		Temperature (°C)			
	total	daily mean	mean	SD	1st	50th	80th	99th
Japan	610613	209	12.9	7.7	-1.9	16.5	24.2	30.6
Hokkaido	26824	9	8.9	6.0	-7.6	9.9	19.7	26.5
Aomori	8644	3	10.8	6.2	-4.9	11.4	20.3	27.6
Iwate	8366	3	11.9	6.4	-5.6	11.0	20.9	27.8
Miyagi	12678	4	11.0	6.9	-1.6	13.7	21.6	28.8
Akita	7026	2	12.1	7.3	-3.2	12.3	22.0	28.7
Yamagata	6907	2	10.0	6.8	-3.5	12.5	22.2	29.0
Fukushima	12676	4	9.4	5.8	-1.5	14.0	22.6	30.0
Ibaraki	15707	5	10.5	6.4	0.6	14.9	22.6	29.3
Tochigi	11601	4	11.7	6.4	0.0	15.2	23.2	29.6
Gumma	11646	4	12.1	8.1	0.7	15.7	24.0	31.0
Saitama	38677	13	12.2	7.0	1.4	16.1	24.2	31.1
Chiba	28992	10	11.5	6.8	2.9	17.1	24.2	30.0
Tokyo	64088	22	14.7	7.8	2.8	17.1	24.3	30.5
Kanagawa	44749	15	14.9	8.0	3.2	17.2	24.1	29.9
Niigata	11826	4	10.2	7.0	-0.4	14.2	23.1	29.7
Toyama	4148	1	9.3	6.5	-0.6	15.3	23.6	30.5
Ishikawa	3891	1	10.3	6.2	0.3	15.7	23.9	30.5
Fukui	2944	1	12.6	7.9	0.1	15.5	24.1	30.6
Yamanashi	5008	2	12.1	6.6	0.2	15.8	24.3	30.1
Nagano	9474	3	10.9	6.5	-3.5	13.1	22.4	28.7
Gifu	12312	4	13.1	7.5	1.1	16.9	25.1	31.7
Shizuoka	18232	6	10.8	6.6	4.0	17.7	24.8	30.0
Aichi	32958	11	14.0	8.0	1.5	16.8	25.1	31.4
Mie	10375	4	13.1	7.9	2.4	16.8	24.8	31.1
Shiga	5617	2	10.5	6.6	0.8	15.5	24.2	30.3
Kyoto	12806	4	12.6	7.0	1.7	16.9	25.2	31.4
Osaka	44485	15	15.4	8.5	2.8	17.5	25.5	31.4
Hyogo	23112	8	13.9	7.3	2.5	17.8	25.3	30.8
Nara	7672	3	11.8	7.1	1.2	16.0	24.3	30.4
Wakayama	4508	2	15.9	8.7	3.0	17.8	25.3	30.6
Tottori	3289	1	13.5	6.7	0.5	15.9	24.0	30.7
Shimane	3752	1	12.4	7.1	0.8	15.8	23.5	30.3
Okayama	7915	3	15.6	8.1	1.4	16.4	24.9	30.9
Hiroshima	9392	3	15.1	8.1	1.6	17.0	25.0	31.1
Yamaguchi	6858	2	15.8	8.6	0.6	16.3	24.2	30.2
Tokushima	2742	1	12.6	7.4	2.8	17.5	25.0	30.5
Kagawa	4430	2	16.2	9.1	2.4	17.2	25.2	31.5
Ehime	8237	3	15.6	7.8	2.5	17.4	25.0	30.5
Kochi	2975	1	13.1	7.0	3.1	18.1	25.1	30.1
Fukuoka	13587	5	16.2	8.9	3.2	18.0	25.1	31.3
Saga	3069	1	14.1	8.1	2.2	17.7	25.3	31.0
Nagasaki	6061	2	13.0	7.0	3.0	18.0	25.0	30.4
Kumamoto	8066	3	16.0	9.0	1.8	17.9	25.5	30.9

Oita	4702	2	13.0	7.1	3.0	17.3	24.6	30.4
Miyazaki	4629	2	12.4	6.5	4.1	18.5	25.4	30.3
Kagoshima	7438	3	17.4	8.0	4.6	19.5	26.5	30.7
Okinawa	5522	2	9.1	5.0	14.0	23.8	28.6	30.5

Table S3. RR₁₁, RR₁₀, RR₀₁, relative excess risk due to interaction (RERI) and 95%CI for the total population and sub-population.

(Sub) groups	RR₁₁	RR₁₀	RR₀₁	RERI
All population	1.04 (1.02,1.05)	1.03 (1.02,1.04)	1.00 (0.97,1.03)	0.01 (0.00,0.02)
Male	1.04 (1.02,1.05)	1.03 (1.01,1.04)	1.01 (0.97,1.06)	-0.01 (-0.02,0.00)
Female	1.02 (1.01,1.03)	1.01 (1.00,1.02)	1.00 (0.96,1.05)	0.01 (0.00,0.02)
≤74years old	1.02 (1.00,1.04)	1.02 (1.01,1.03)	1.00 (0.96,1.04)	0.02 (0.01,0.03)
>74years old	1.04 (1.03,1.05)	1.03 (1.02,1.04)	1.01 (0.97,1.05)	0.00 (-0.01,0.01)
Adj. humidity	1.04 (1.02,1.05)	1.03 (1.02,1.04)	0.99 (0.96,1.03)	0.01 (0.00,0.02)
Lag 0-1	1.03 (1.01,1.04)	1.02 (1.01,1.03)	1.01 (0.98,1.04)	0.00 (-0.01,0.01)

RR₁₁: PM_{2.5} effect when temperature is centered at 99th percentile. RR₁₀: PM_{2.5} effect when temperature is centered at 80th percentile. RR₀₁: Heat effect (99th vs 80th temperature percentile) when PM_{2.5} concentration is centered at 5μg/m³. RERI=RR₁₁-RR₁₀-RR₀₁+1. To obtain the CI of RERI, RERI was estimated 1000 times by bootstrap replications (with replacement) of the original sample. Bold fonts for RERI represent 95% CIs > 0.

In the sensitivity analysis, we did not calculate the RERI for the six warmest months because the temperature distribution changed when we limited the study period to these months. The reference point for the heat effect was the 80th temperature percentile, making the RERI from 6 warmest months incomparable with those of the other groups.

Table S4. The attributable number (AN) and attributable fraction (AF) to co-exposure and single exposure of temperature and PM_{2.5}. Due to missing PM_{2.5} data in some prefectures, days without PM_{2.5} data and do not meet the exposure criteria were excluded from the analysis assessing the co-exposure of PM_{2.5} and heat. After excluding periods without PM_{2.5} data, the total number of cardiac-origin OHCA was 577,393.

Exposure (criteria)	AN	AF(%)	Exposure days (% from study period)
PM _{2.5} (Temperature per>80 th & PM _{2.5} >5μg/m ³)	1276	0.22	24786 (20%)
Heat (Temperature per>80 th)	1215	0.21	24786 (20%)
Co-exposure of PM _{2.5} and heat	2638	0.46	24786 (20%)

Table S5. Sensitivity analysis of the independent associations between daily mean air temperature, PM_{2.5}, and OHCA risk.

Exposure	Model	RR	95%CI
Heat effect (99th vs 80th)	Main model	1.04	(1.02-1.06)
	Adj humidity	1.03	(1.01-1.05)
	6 warmest months	1.09	(1.06-1.12)
PM _{2.5} effect (every 10µg/m ³ increase)	Main model	1.017	(1.010-1.024)
	Adj humidity	1.017	(1.010-1.025)
	6 warmest months	1.020	(1.010-1.031)

Figure S1. The pooled results for OHCA relative risk for every 10 $\mu\text{g}/\text{m}^3$ increase in PM_{2.5} concentrations at different lag periods.

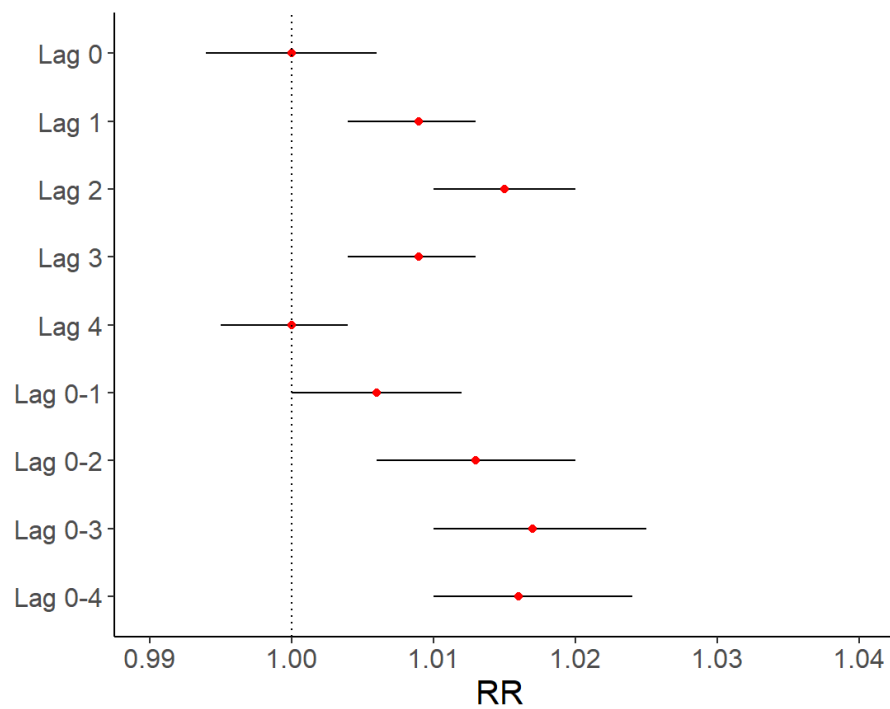


Figure S2. Boxplots (median, first, and third quartiles are shown as horizontal lines and outliers as dots) of daily mean PM_{2.5} concentrations in 47 Japanese prefectures from 2012 to 2019.

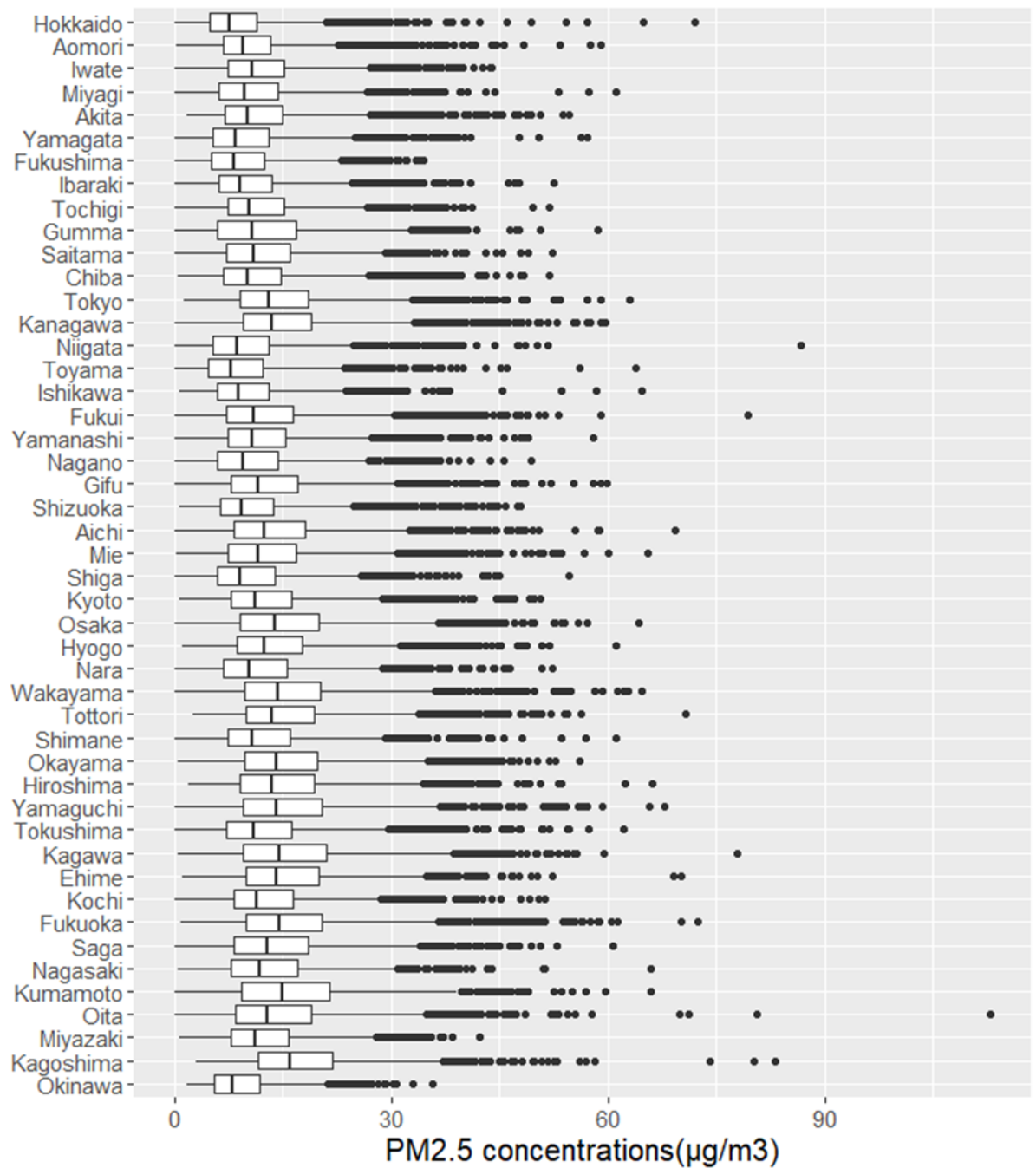


Figure S3. Pooled association between PM_{2.5} concentration and morbidity across 47 Japanese prefectures using a spline model. In the spline model, a non-linear PM_{2.5} effect was modeled using a cross-basis function of DLNM applied on daily PM_{2.5} concentration, with lag periods up to 7 days. The PM_{2.5} concentration-morbidity curve was modelled with a quadratic B-spline function with 2 knots at 50th and 90th percentiles. The lag-response curve was also modelled with a natural cubic spline with 2 internal knots placed at lag 1 and lag 3. The curve is centered at 6 μ g/m³.

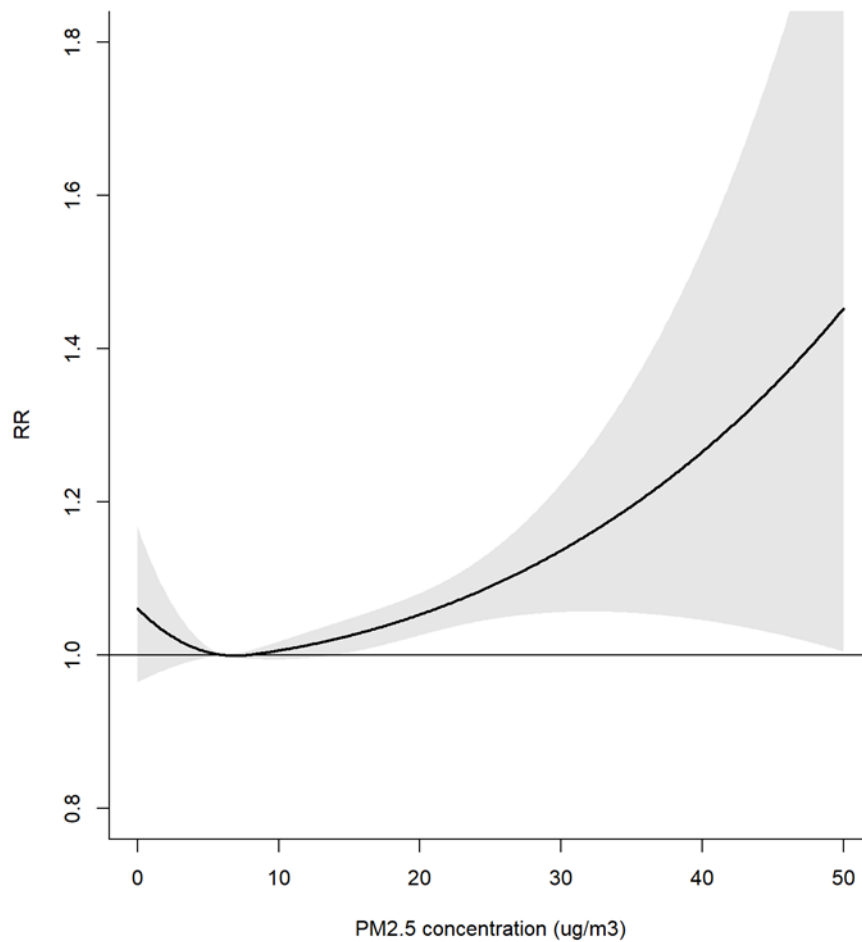


Figure S4. The dataflow of OHCA cases in Japan from 2012 to 2019. VF stands for ventricular fibrillation.

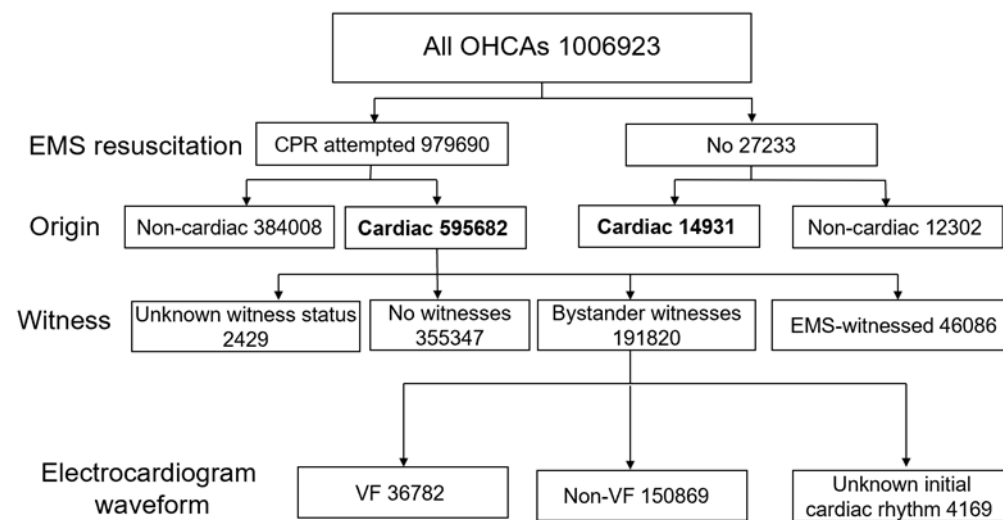


Figure S5. Time series plot of daily cardiac-origin OHCA cases in Japan from January 2012 to December 2019.

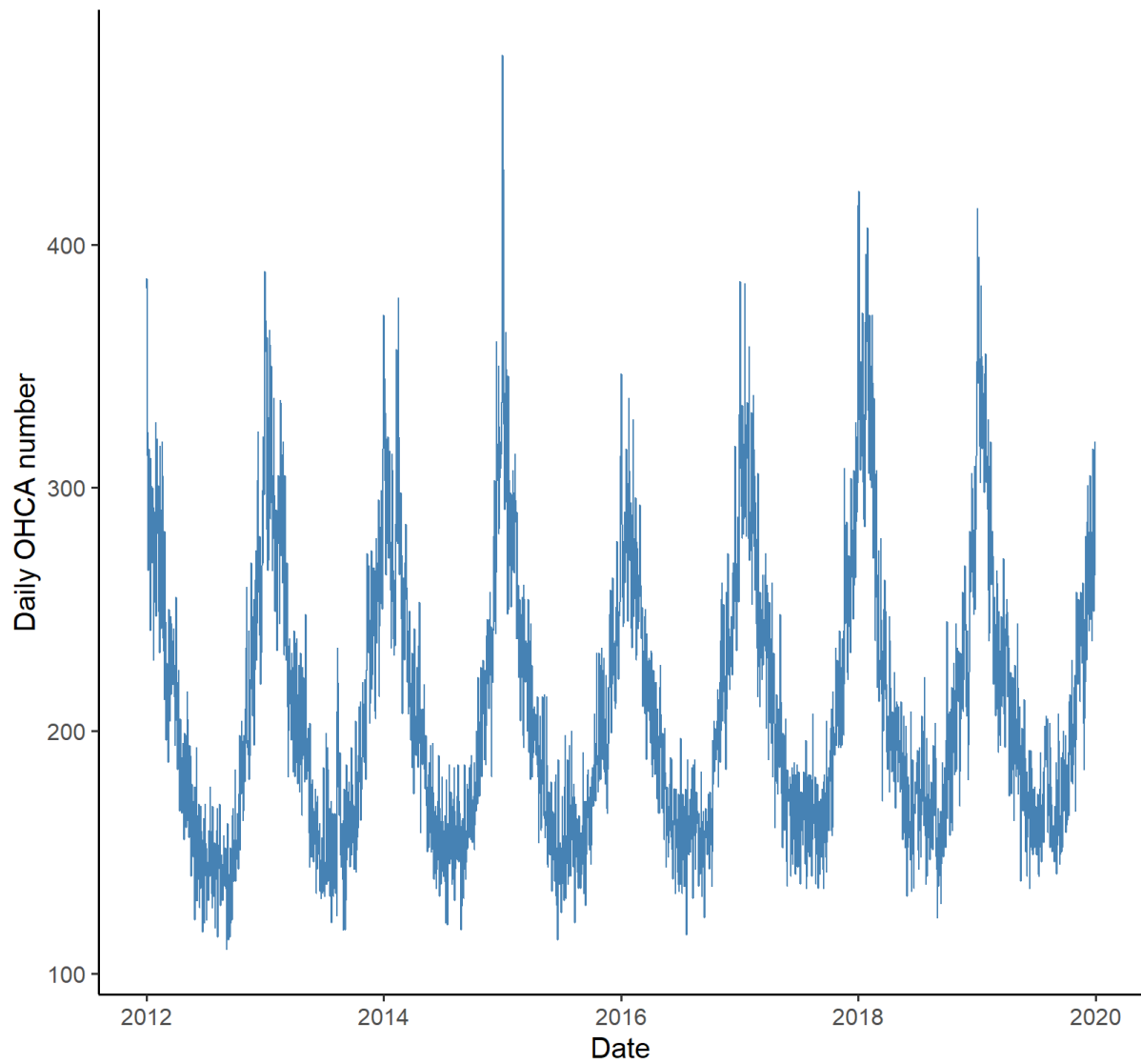


Figure S6. Boxplots (median, first, and third quartiles are shown as horizontal lines and outliers as dots) of daily mean temperature in 47 Japanese prefectures from 2012 to 2019.

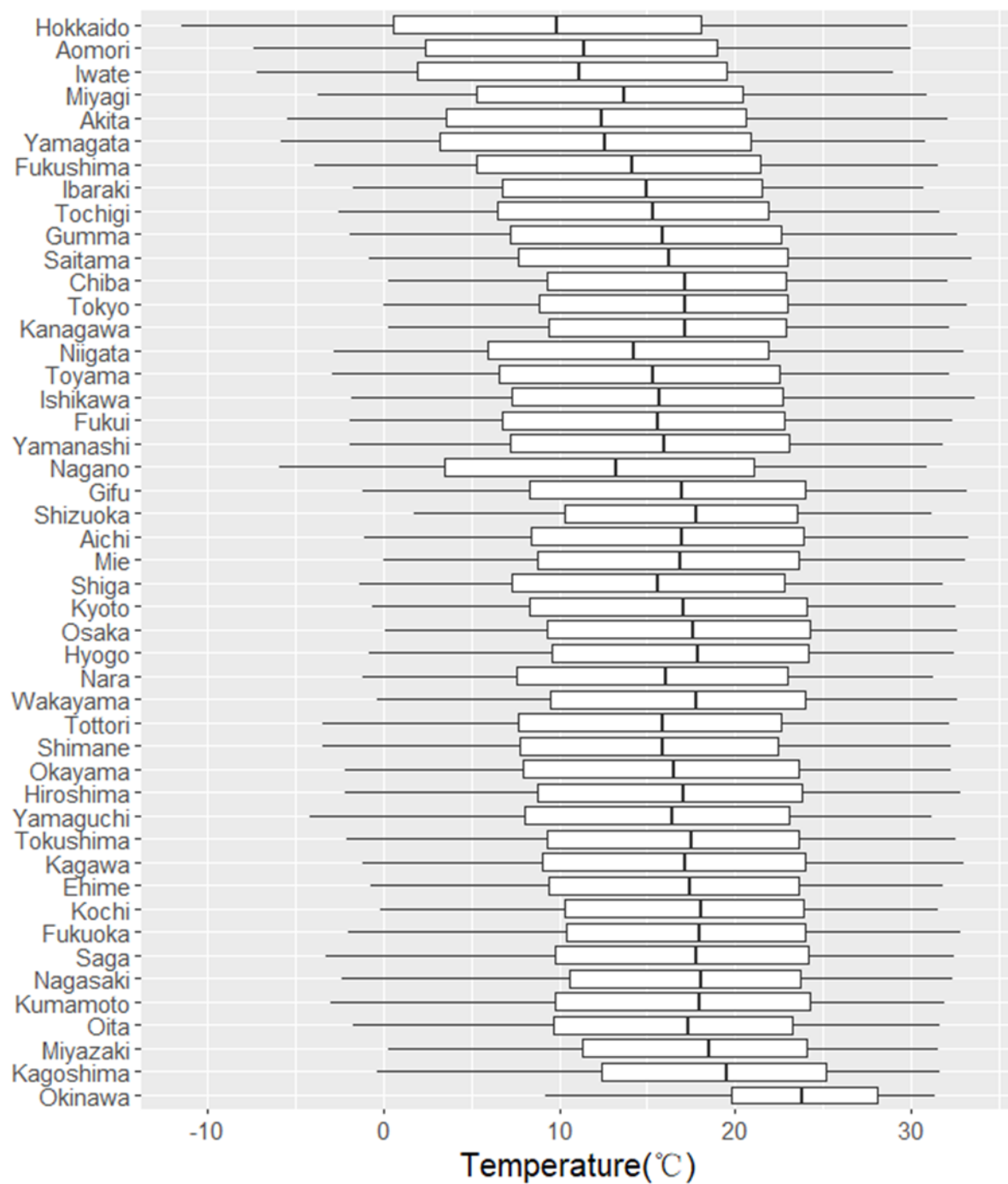


Figure S7. Boxplots (median, first, and third quartiles are shown as horizontal lines and outliers as dots) of daily mean temperature in each month from 2012 to 2019 in Japan.

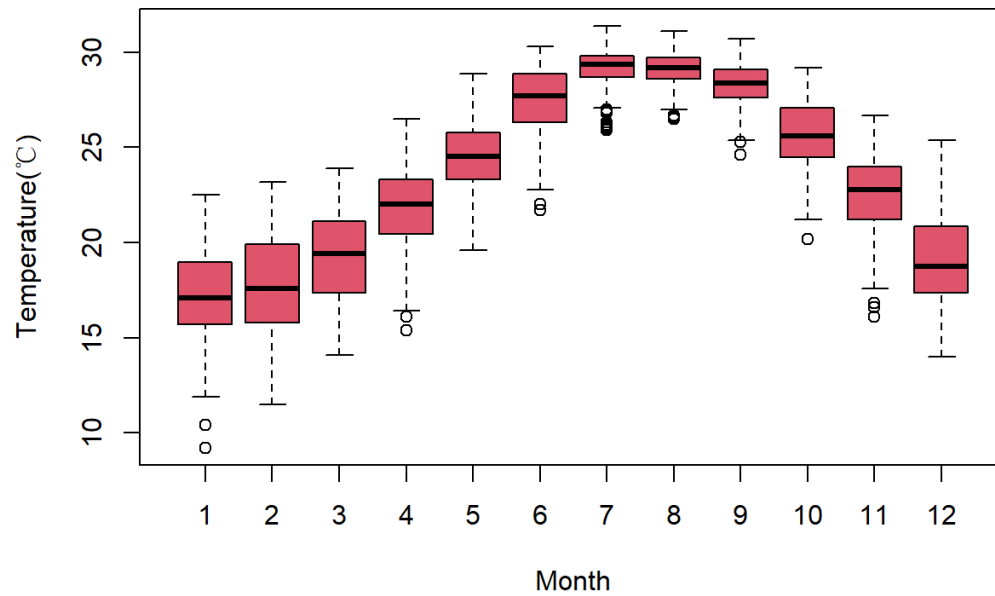


Figure S8. Pooled temperature–OHCA risk curves at 1 $\mu\text{g}/\text{m}^3$ and 40 $\mu\text{g}/\text{m}^3$ $\text{PM}_{2.5}$ across 47 Japanese prefectures.

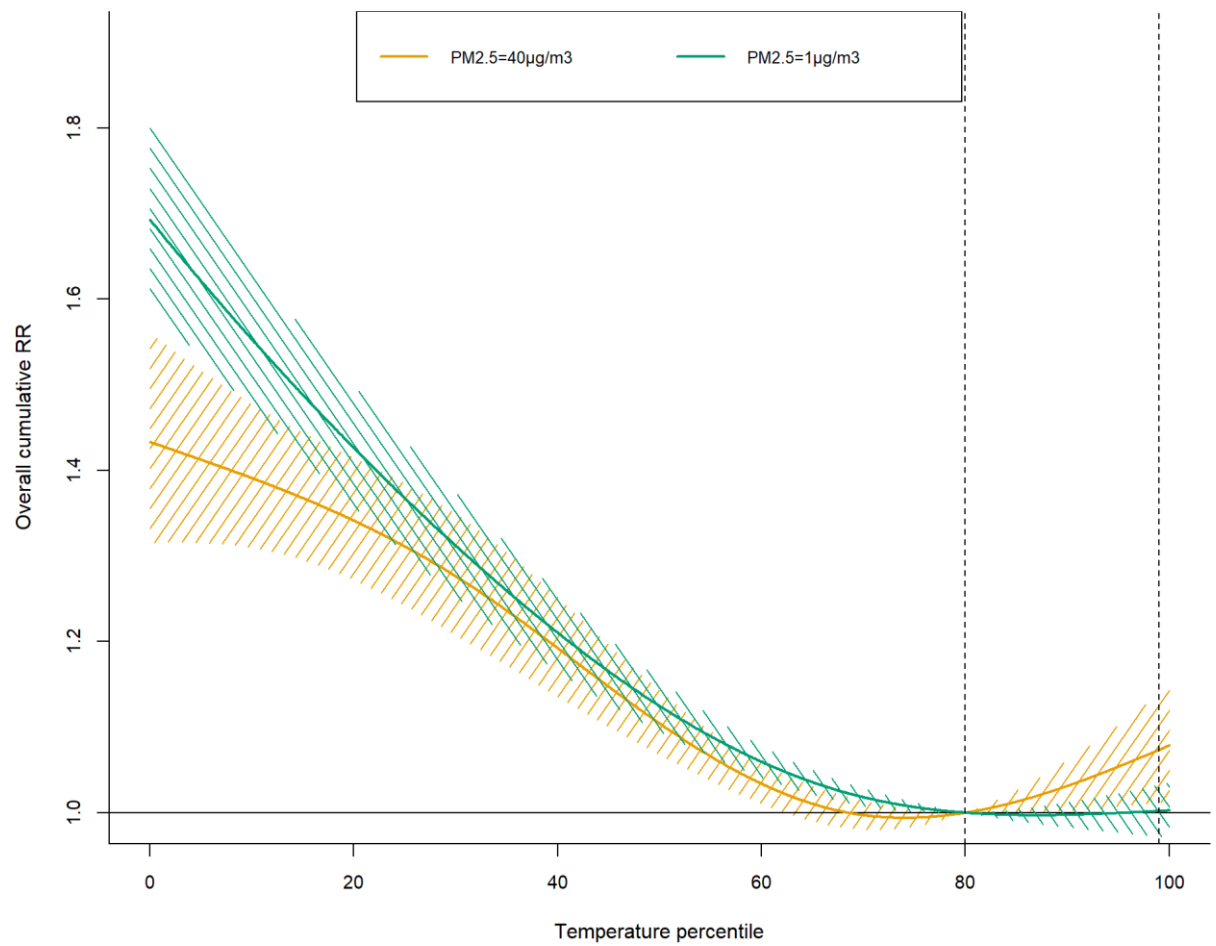


Figure S9. Pooled $\text{PM}_{2.5}$ concentration–specific heat effects (RR at 80th vs 99th temperature percentile) across 47 Japanese prefectures at $\text{PM}_{2.5}$ levels from 1 to $40 \mu\text{g}/\text{m}^3$, by region: (a) All prefectures; (b) Excluding Hokkaido; (c) Excluding Kyushu and Okinawa; (d) Kyushu and Okinawa only.

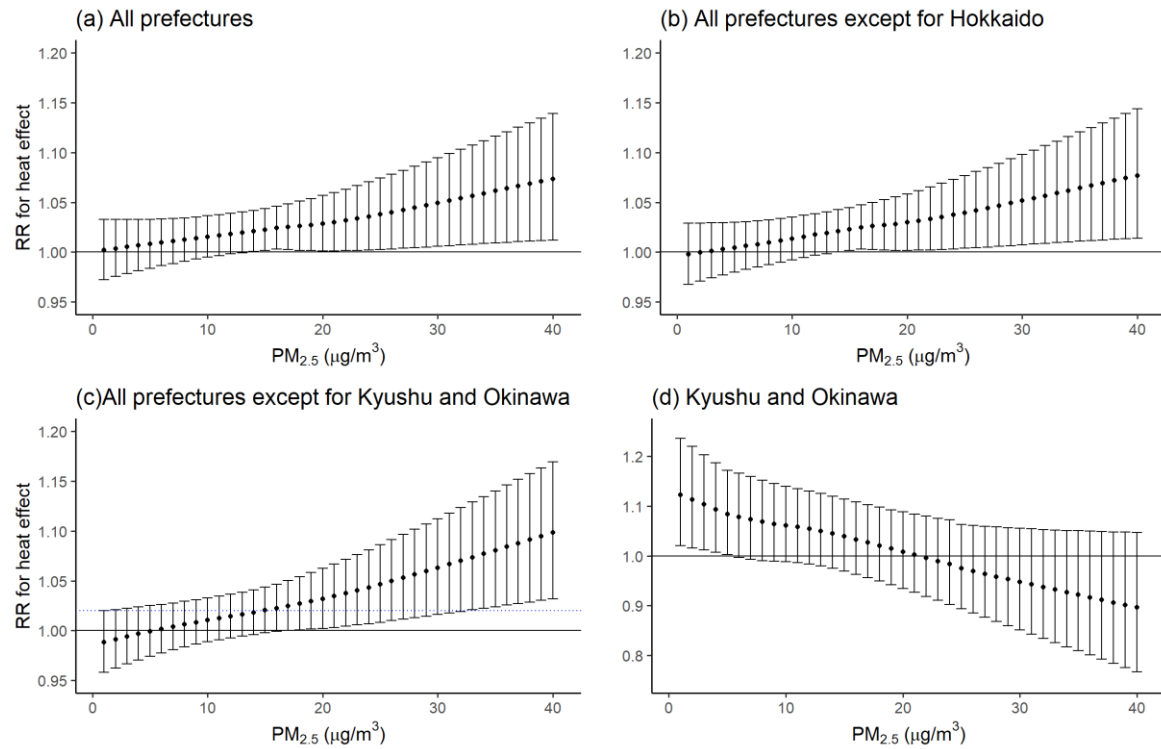


Figure S10. Pooled temperature-specific PM_{2.5} effects (RR per 10 µg/m³ increase) across 47 Japanese prefectures at temperature percentiles, by region: (a) All prefectures; (b) Excluding Hokkaido; (c) Excluding Kyushu and Okinawa; (d) Kyushu and Okinawa only.

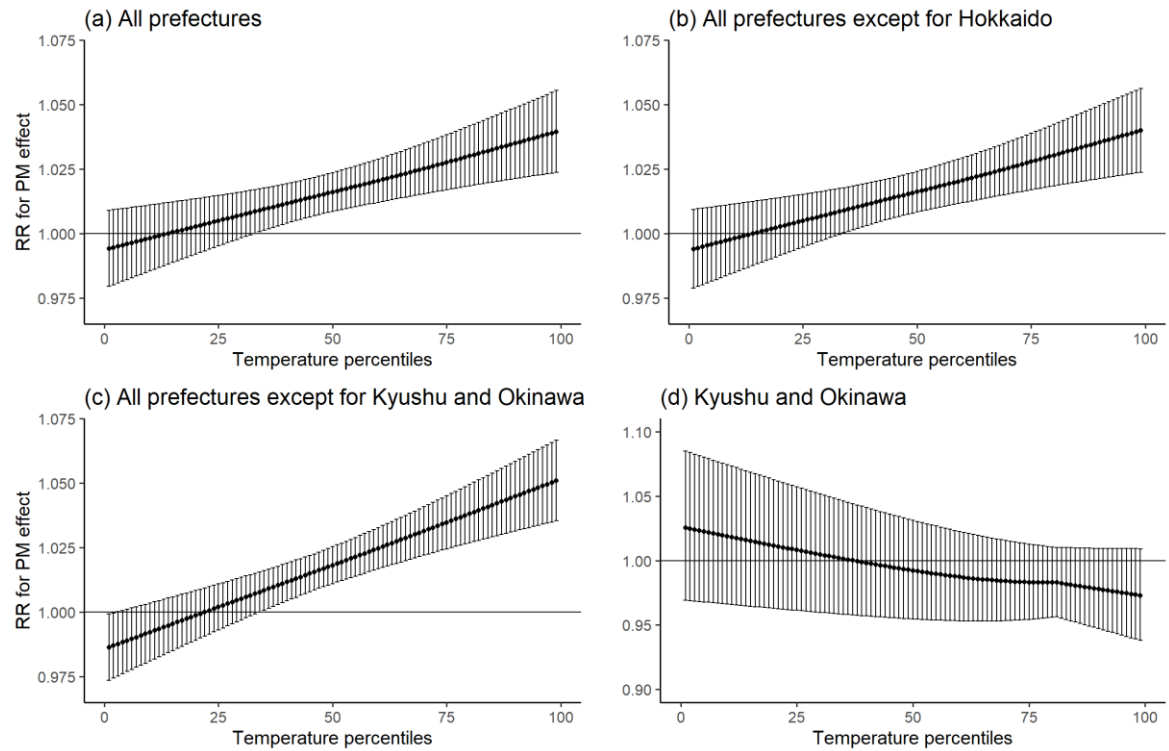


Figure S11. Pooled PM_{2.5} concentration–specific heat effects (RR at 80th vs 99th temperature percentile) across 47 Japanese prefectures for sensitivity analyses: (a) Main model; (b) Adjusted for humidity; (c) Six warmest consecutive months; (d) Lag 0–1.

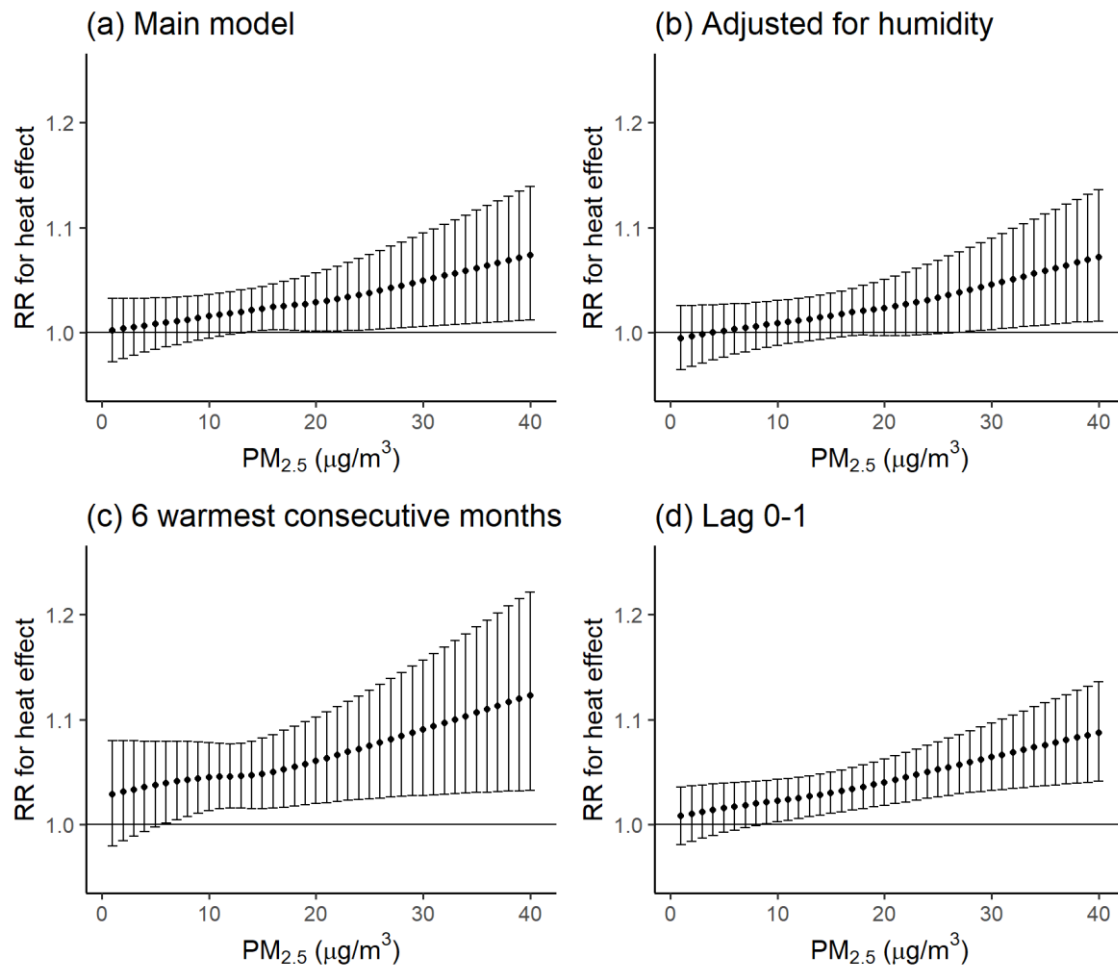


Figure S12. Pooled temperature-specific PM_{2.5} effects (RR per 10 µg/m³ increase in PM_{2.5} concentration) across 47 Japanese prefectures for sensitivity analyses: (a) Main model; (b) Adjusted for humidity; (c) Six warmest consecutive months; (d) Lag 0–1.

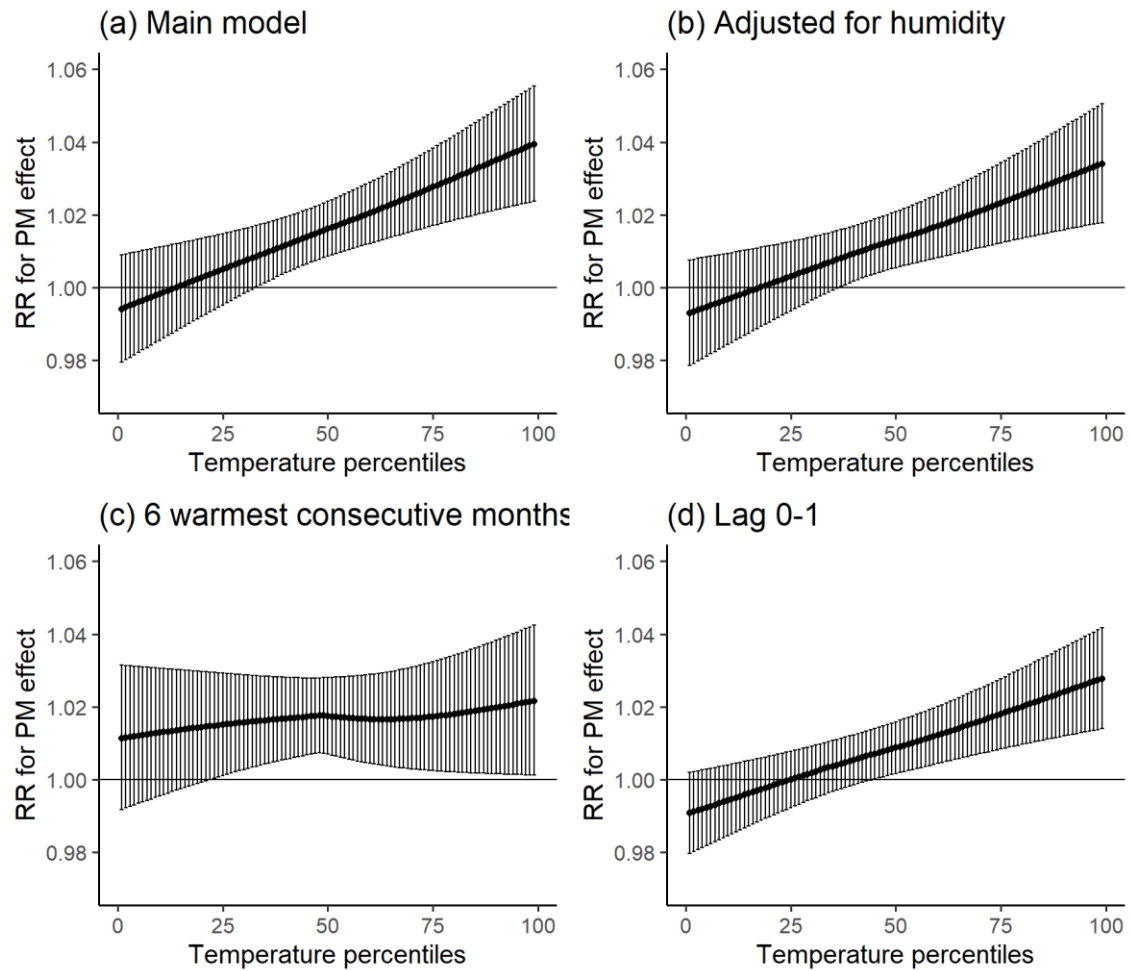


Figure S13. Pooled $\text{PM}_{2.5}$ concentration-specific heat effects (RR at 80th vs 99th temperature percentile) for target populations: (a) cardiac-origin OHCA; (b) bystander-witnessed OHCA; and pooled temperature-specific $\text{PM}_{2.5}$ effects (RR per $10 \mu\text{g}/\text{m}^3$ increase) for (c) cardiac-origin OHCA; (d) bystander-witnessed OHCA.

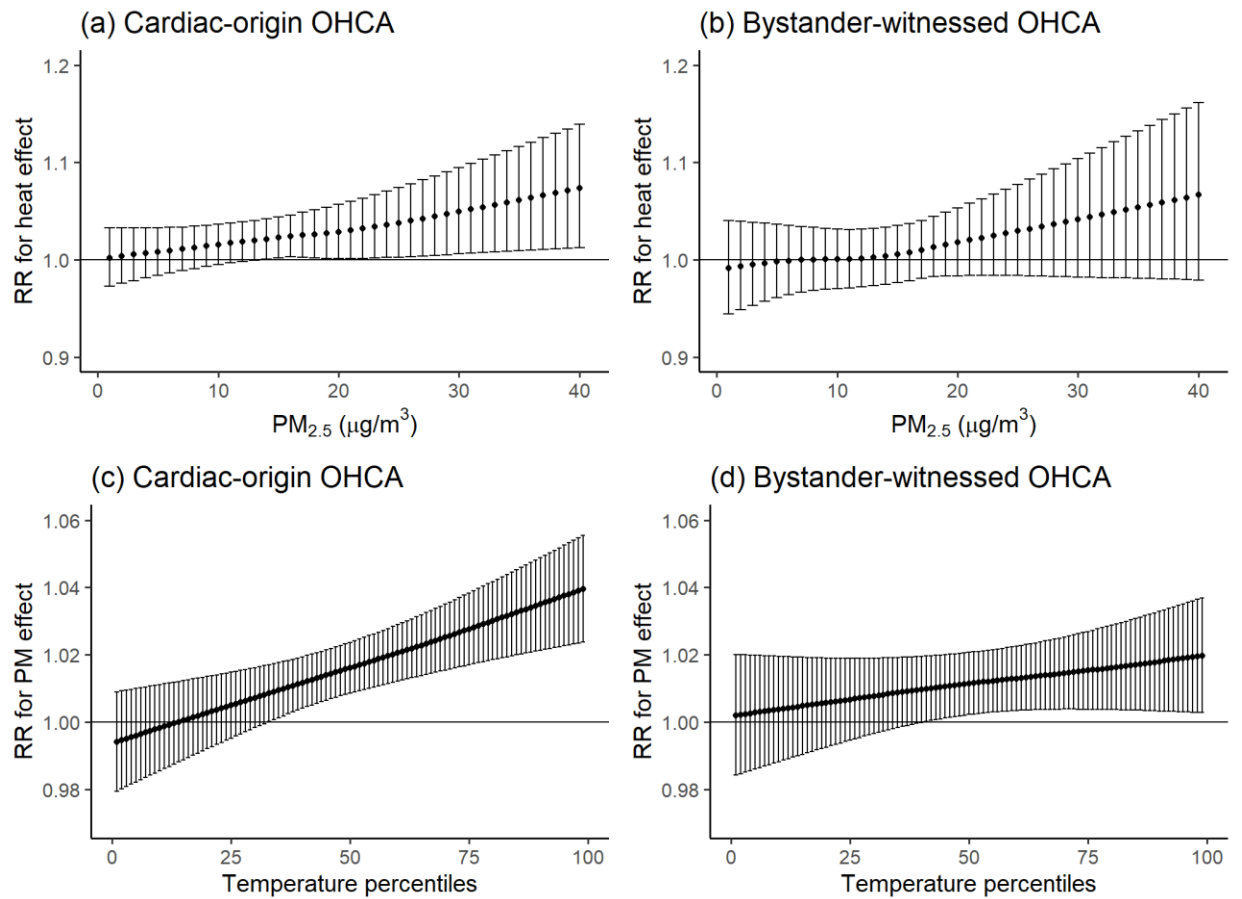


Figure S14. Pooled PM_{2.5} concentration–specific cold effects (RR at 1st vs 80th temperature percentile) for target populations: (a) Main model; (b) Adjusted for humidity.

